Best Practices for Biomedical Research Data Management

Module 6: Biomedical Ethics

IACUC: Institutional Animal Care & Use Committee

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This video will provide an example of an Institutional Animal Care and Use Committee (IACUC) in ensuring the ethical and sensitive care and use of animals in research, teaching and testing. Every institution that uses animals for federally funded laboratory research must have an Institutional Animal Care and Use Committee.

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What is an IACUC?

- Local institutional committee with federally mandated oversight responsibility on animal use
  - Review and approve animal use protocols
  - Monitor institutional animal care and use programs
  - Support animal users and institutions

What is Animal Use Activity?

- Use vertebrate animals in
  - Research
  - Teaching and Training
  - Testing
  - Generation of custom antibodies

U.S. regulations and requirements

- The Animal Welfare Act
  - All warm-blooded animals except rats, mice and birds, and farm animals used in food and fiber research
- Public Health Service (PHS) Policy
  - Research institutions that receive support through the United States PHS for research involving animals
- Guide for the Care ad Use of Laboratory Animals
- Guide for the Care and Use of Agricultural Animals in Research and Teaching

Which animals may require IACUC review?

- Vertebrate animals include:
  - Laboratory animals
  - Client owned animals
  - Farm animals
  - Wildlife
  - Aquatic animals
- *This may be depending on the institution*

Investigator Responsibilities

- Investigators are accountable for the protection of the research and teaching animals
  - Obtain IACUC approval prior to using animals
  - Conduct research according to the approved protocol
  - Comply with policies and procedures

http://grants.nih.gov/grants/olaw/InvestigatorsNeed2Know.pdf

Animal Use Protocol

- Binding
- Maximum interval between IACUC review and approval is three years
- Protocol can be modified
- USDA and Department of Defense require annual review
- Two valid methods by the IACUC
  - Full committee review
  - Designated member review

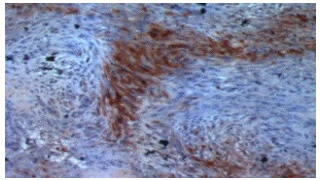
Obtaining IACUC Approval

- Animal use protocols require to describe:
  - Rational for animal use and consideration of alternatives
  - Species and animal number justifications
  - Procedures, endpoint and euthanasia method
  - Methods to minimize pain and distress
- Find resources early
- Time consideration

Consideration: 3 Rs

- **Replacement**: substitute animals with non-animal methods
- **Reduction**: minimize the number of animal use
- **Refinement**: avoid or minimize discomfort, distress and pain
  - Less invasive procedures
  - Use of analgesia

Number Reduction



- Dr. A wants to follow the gene expression in the brain using a special dye
- Old method: animals need to be euthanized for *in vitro* analysis at 3 time points requiring 30 animals (10 animals x 3)
- New method: non-invasive live imaging allows sequential imaging using the same animals requiring 10 animals

Minimizing Pain and Distress

- Dr. C wants to test therapeutic effects of the agent in the fracture animal model
- Both legs: right-fracture with the agent and left-fracture without
- One leg: either fracture with or without the agent
- Double the animal number, but less pain and distress

Common Compliance Issues

- Absence of IACUC approval
- Deviation from the approved protocol
- Incomplete documentation
- Luck of Training
- *Suspension of animal activity by the IACUC will be reported to federal agency and funding sources*

Applying for grants

- Protocol and grant congruency evaluation
- Internal collaborator’s animal protocols
- External collaborator's animal protocols
- Service agreement
- Memorandum of Understanding

How would you handle this call to the IACUC office?

- Caller: My research adviser said I need to contact the IACUC office and see if I need IACUC approval. I'm planning to obtain discarded mouse tissues to extract 100 proteins.

Where can I get help?

- IACUC office, Office of Research
  - Any questions
  - Protocol development
- Veterinary Service
  - Consultation
  - Hands on training
- Library
  - Alternative search
  - Managing animal records
  - General data management

Resources

- Guide for the Care ad Use of Laboratory Animals
  https://grants.nih.gov/grants/olaw/Guide-for-the-Care-and-use-of-laboratory-animals.pdf
- Guide for the Care and Use of Agricultural Animals in Research and Teaching
  http://fass.org/docs/agguide3rd/Ag_Guide_3rd_ed.pdf
- AVMA Guidelines for the Euthanasia of Animals: 2013 Edition
  https://www.avma.org/KB/Policies/Documents/euthanasia.pdf
- Institutional Animal Care and Use Committee Guidebook
  http://grants.nih.gov/grants/olaw/GuideBook.pdf
- Office of Laboratory Animal Welfare (OLAW)
  http://grants.nih.gov/grants/olaw/olaw.htm
- USDA Animal and Plant Health Inspection Service (APHIS)
  http://www.aphis.usda.gov/